Powering ArcGIS Apps with R

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ESRI DEVELOPER & TECHNOLOGY SUMMIT

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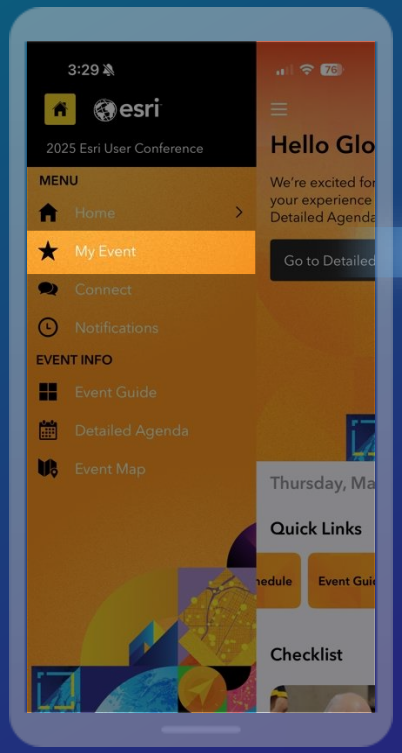
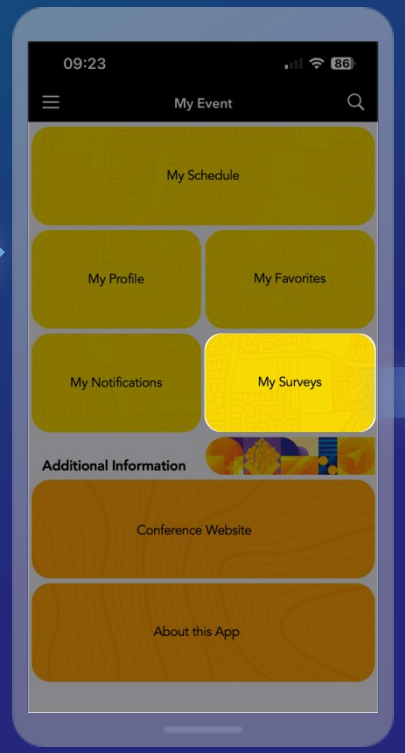
Please share your feedback in the app

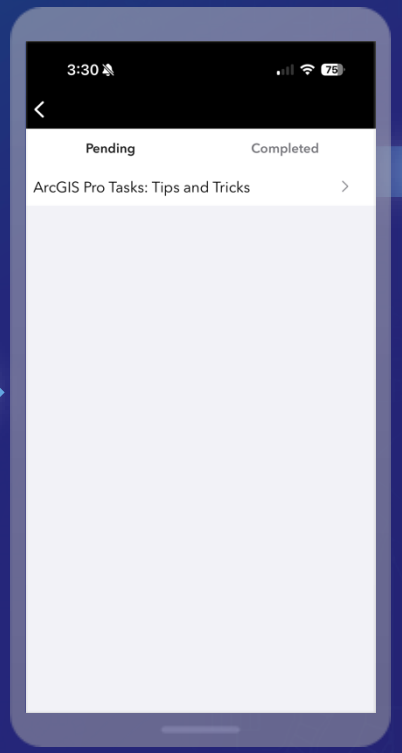
Select “My Event” in the event app main menu



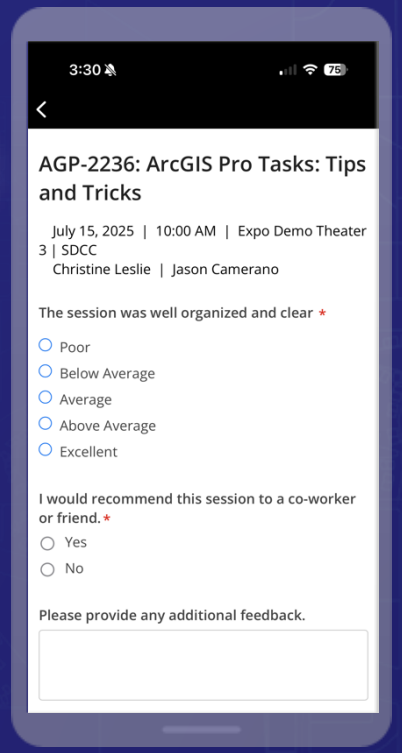
Select “My Surveys”

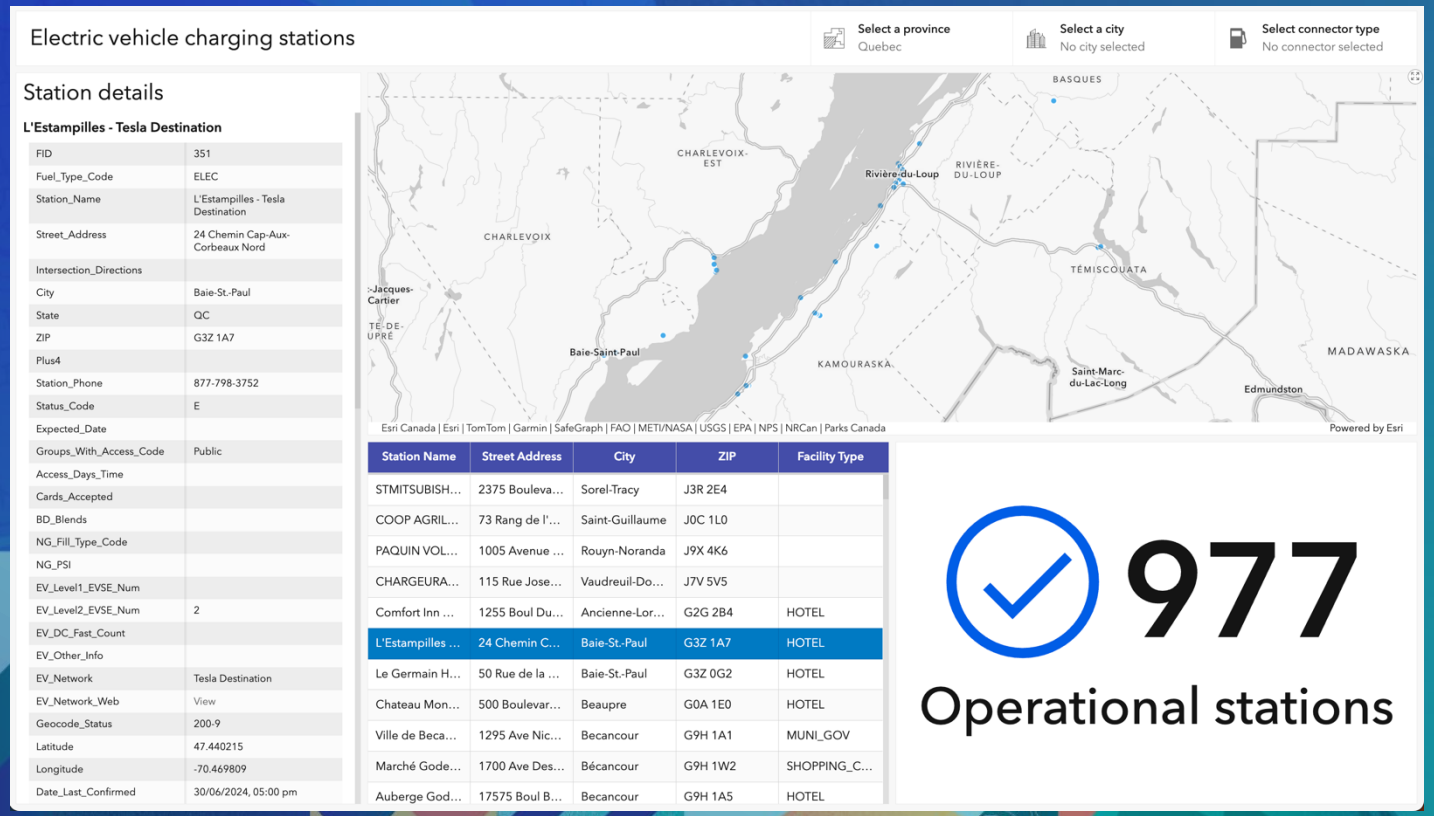


Select the desired survey



Complete the survey





Dashboards backed by R

What is the R-ArcGIS Bridge?

R-ArcGIS Bridge

+

ArcGIS

R-ArcGIS Bridge

- Collection of R packages
- Integrates the R language with the ArcGIS ecosystem
- Open source
- No license required!

What is your environment & use case?

Esri components	*R package*	*Use case*
• Location and Data Services • ArcGIS Online • ArcGIS Enterprise • ArcGIS Platform	**{arcgis}**	• Read/publish/edit a feature or image service • Read Survey123 data • Download attachments • Use a geocoding service • Build an app with calcite
• ArcGIS Pro geoprocessing framework - ArcToolbox - GP Services • Geodatabases	**{arcgisbinding}**	• Publish a geoprocessing service • Create and share an ArcToolbox driven by R scripts • Read/write to file gdb

ArcGIS Packages

- {arcgislayers}
- {arcgisutils}
- {arcgisgeocode}
- {calcite}

{arcgislayers}

- Connect to any ArcGIS Data Service
  - Enterprise
  - Hosted
  - Survey123
- Publish your R objects
- Modify data in place
- Attachments:
  - Query
  - Download
  - Update Attachments

Access your data, anywhere

{arcgisutils}

- 🔐 **Comprehensive Authentication**
  - OAuth2
  - API Keys
  - Shiny integration
- 🌐 **Portal Integration**
  - **Users and Groups**
  - **Portal Item access**
  - **Search**
- 🛠️ Geoprocessing Services
- 👩🏻‍💻 Developer Utilities

```r
library(arcgisutils)

# authenticate
set_arc_token(auth_user())

# store the service url
service_url <- "https://hydro.arcgis.com/a

# create a new job
downstream_job <- arc_gp_job$new(
  base_url = service_url,
  params = list(InputPoints = as_esri_featureset(input_points), f =
  # the result is a GPRecordFeatureSet
  result_fn = parse_gp_feature_record_set,
  token = arc_token(),
)

# print the job
downstream_job
```

```
#> <arc_gp_job>
#> Job ID: not initiated
#> Status: not started
#> Resource: /TraceDownstream
#> Params:
#> • InputPoints
#> • f
```

{arcgisgeocode}

Fast Geocoding built in Rust

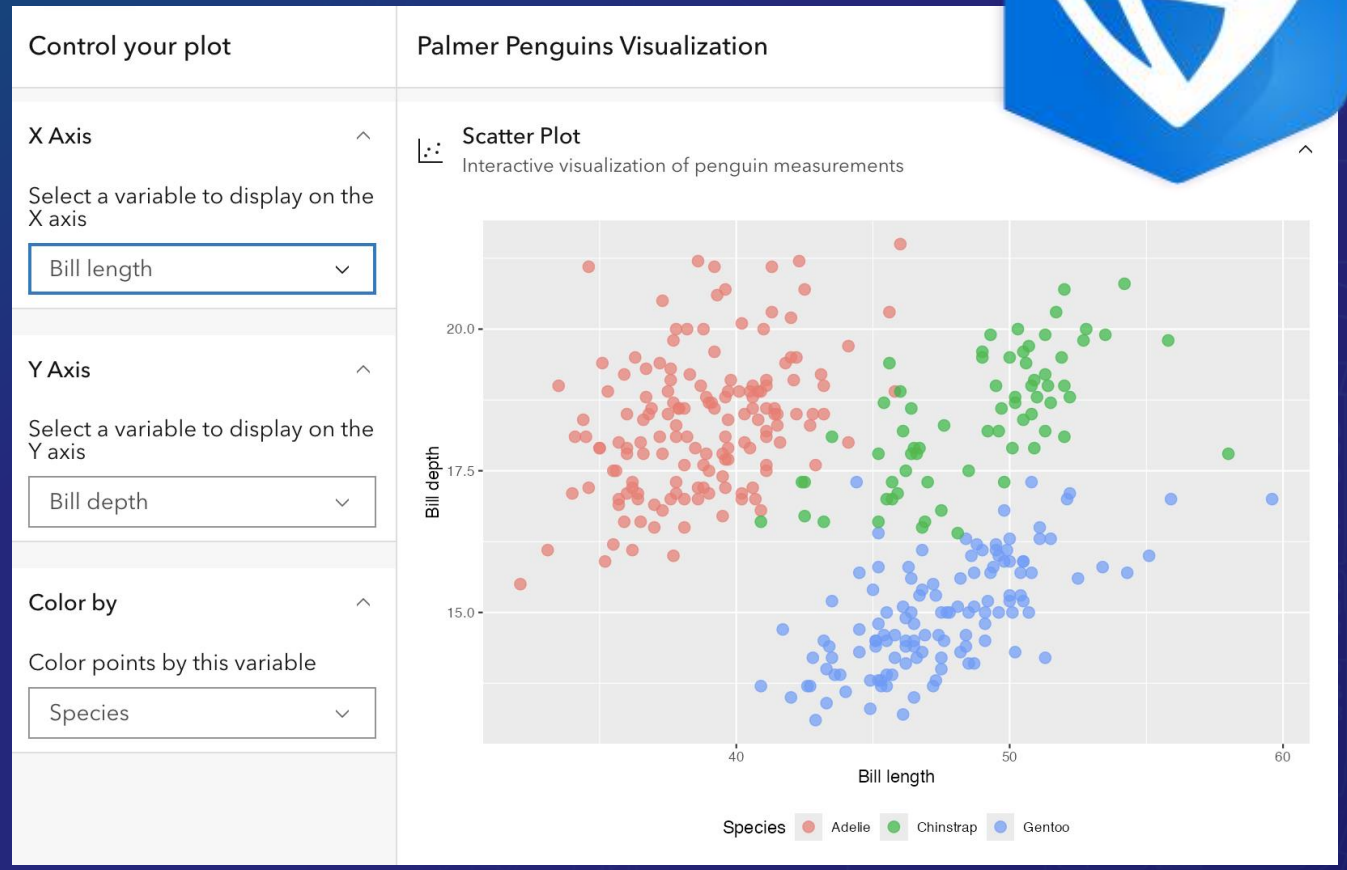
- Custom Locators (e.g. StreetMap Premium)
- Parallel Processing

- **Capabilities**
  - Address Candidate Search
  - Reverse Geocoding
  - Batch Geocoding
  - Interactive Suggestions

{calcite}

Create beautiful apps UIs 👩‍🎨

- {shiny} integration
- Create app layouts
- Pre-built inputs & components
- Same library that ArcGIS Online and the ArcGIS Maps SDK uses!



R-ArcGIS Bridge w/ 🤖 AI Assistants

llms.txt – Context for your agents 🤖 😅

Context for Claude, Codex, Gemini, and more

- {arcgislayers}:
  - https://r.esri.com/arcgislayers/llms.txt
- {arcgisutils}:
  - https://r.esri.com/arcgisutils/llms.txt
- {arcgisgeocode}:
  - https://r.esri.com/arcgisgeocode/llms.txt
- {calcite}:
  - https://r.esri.com/calcite/llms.txt

arcgis

R

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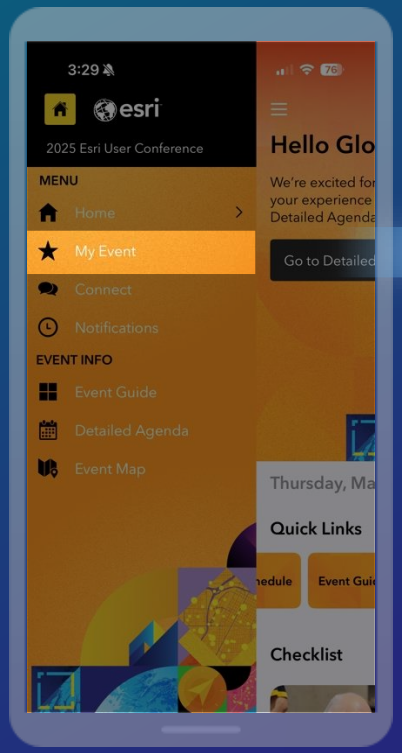

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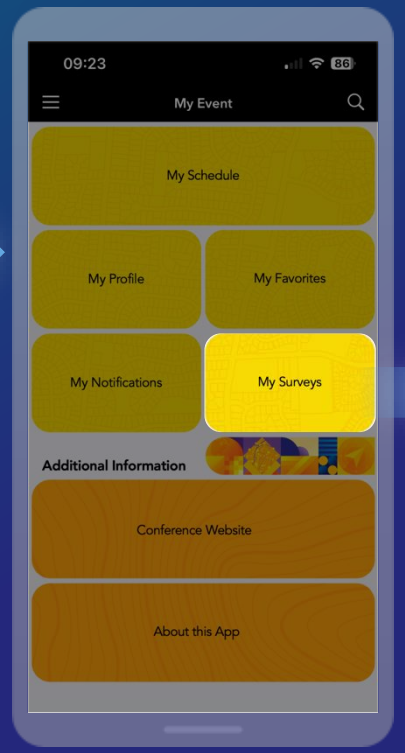
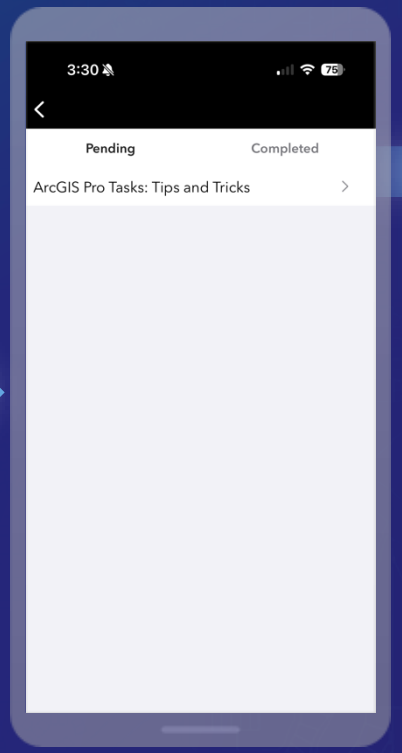
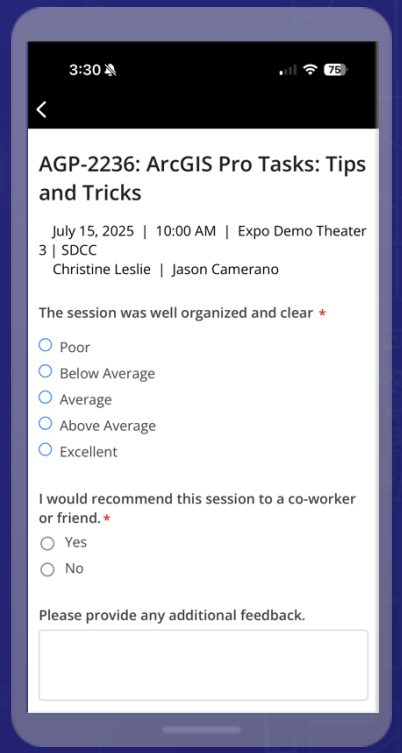


Select the desired survey



Complete the survey



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Questions

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[Add section title]

Agenda

- [Add section title]
- [Add section title]
- [Add section title]
- Questions

[Add section title]

[Add topic]

- [Add subtopic]
- [Add subtopic]
- [Add subtopic]

[Add topic]

[Add big word]

[Add smaller word]

[Add “Quote”]

[Name]

Demo: [Add demo title]

[Add topic]

- [Add subtopic]
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- `[Add code]`
  - [Add description]

- [Add subtopic]
  - `[Add code]`

[Add topic]

- [Add subtopic]
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Acknowledgments

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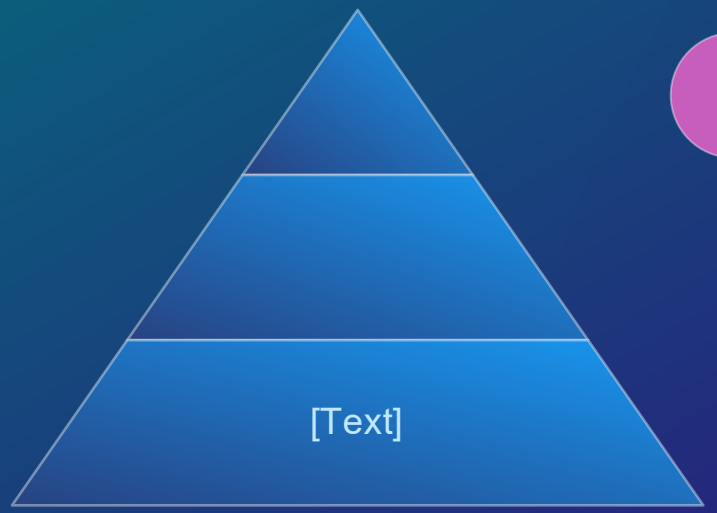
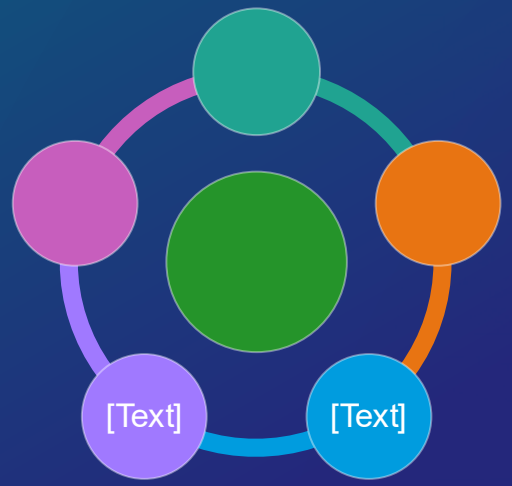
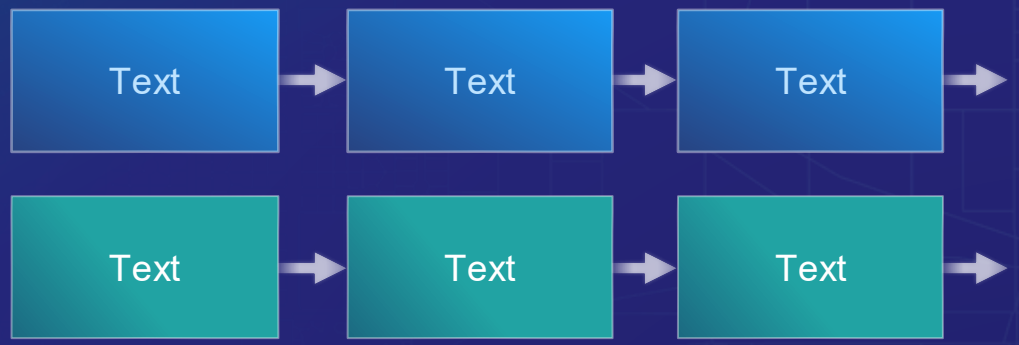
Source: [add source information]

Color suggestions







Use copy/paste formatting
to apply these color options

Solids

Gradients

Code color suggestions
Description of developer code `[insert code]`

For additional diagram options, see the *Diagrams, Tables, and Timelines template*

What is [add product name, capability, or technology]?

- [Add simple description of product, capability, or technology]
  - [Add capability]
  - [Add capability]
  - [Add capability]

What is [add product name, capability, or technology]?

- [Add description]
- [Add description]
- [Add description]

[Add product or solution name] road map

Near-term	Midterm	Long-term
• [Add enhanced functionality in development] • [Add enhanced functionality in development]	• [Add enhanced functionality in development] • [Add enhanced functionality in development]	• [Add enhanced functionality in development] • [Add enhanced functionality in development]

Speaker introductions

[Add name of speaker]

[Add name of speaker]

[Add name of speaker]

[Add name of speaker]

[Add product name] vision and strategy

- [Add subtopic]
- [Add subtopic]
- [Add subtopic]

What is [add feature]?

- [Add simple description of application or solution feature]
  - [Add capability]
  - [Add capability]
  - [Add capability]

[Add topic]

Currently available	Changes coming
• [Add existing functionality A] • [Add existing functionality B]	• [Add enhanced functionality A in development] • [Add enhanced functionality B in development]

[Add topic]

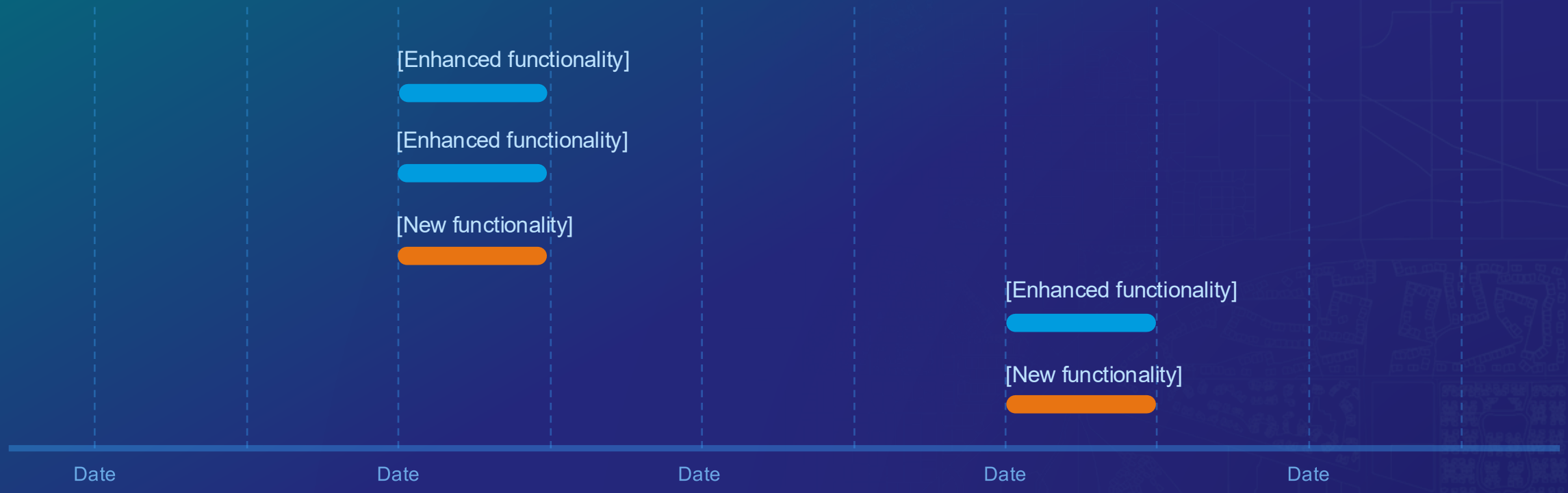
[Add optional caption]

[Add optional caption]

[Add product or solution name] road map

Near-term	Midterm	Long-term
• [Add enhanced functionality in development] • [Add enhanced functionality in development]	• [Add enhanced functionality in development] • [Add enhanced functionality in development]	• [Add enhanced functionality in development] • [Add enhanced functionality in development]

[Add product name] release schedule



What is [add product name]?

- [Add simple description of application or solution type]
  - [Add capability]
  - [Add capability]
  - [Add capability]

Highlights

- [Add highlight]
- [Add highlight]
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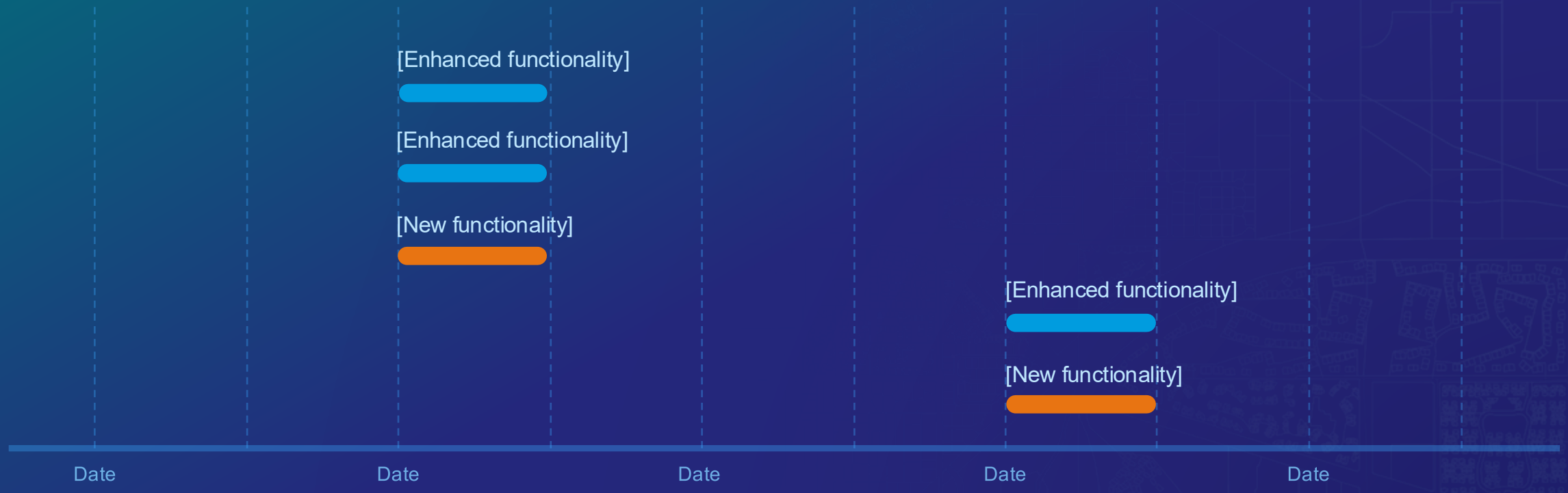
[Add product name] release schedule